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Sustainable Solutions for Marginal Fields: Cost-Effective Design Strategies for Onshore Surface Facilities

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Abstract

Marginal fields, often characterized by limited reserves and low production rates, pose unique challenges in terms of return on investment (ROI) and long-term sustainability. Rising global energy demand and stringent environmental standards necessitate innovative and cost-effective approaches to field development. This paper introduces a holistic framework for designing Onshore Surface facilities that integrates combining lean engineering, modular design, digitalization, fiscal management, cost-effective execution, risk management, and economic efficiency. Drawing upon the successful case study of the Haliba Field by Al Dhafra Petroleum (ADNOC), the paper outlines practical pathways for transforming marginal fields into economically viable and environmentally sustainable energy assets. These strategies collectively aim to redefine marginal field economics through replicable, efficient, and environmentally conscious design models.

Introduction

Background

Al Dhafra Petroleum Operations Company Limited (ADNOC), a joint venture with KADOC established in late 2013, holds onshore concession areas in Abu Dhabi, UAE. The company currently operates the Haliba field, which serves as a central production gathering hub for hydrocarbons from Haliba and adjacent satellite fields. Produced well-fluids are routed to the Haliba Central Gathering Facilities (CGF) via Multi-Port Selector Valves (MPSVs). These valves enable selective routing from multiple wells (up to seven per MPSV), optimizing flow control and testing operations. At the CGF, fluids are boosted using Multi-Phase Pumps (MPPs) and transferred through a ~65 km pipeline to ADNOC Onshore ASAB facilities for downstream processing and export.

To address metering uncertainties associated with Multi-Phase Flow Meters (MPFMs), Custody Transfer Metering Facilities (CMF) have been installed at Haliba CGF. These facilities, operated by OILSERV

(DBO Contractor), include a 3-phase separator, oil booster pump, and metering skid, ensuring accurate phase separation and fiscal measurement before recombination at the MPP suction for onward transfer. This infrastructure enables Al Dhafra Petroleum to leverage existing ASAB processing capabilities while maintaining operational efficiency and measurement integrity across its production network.

Marginal fields, typically defined as fields with recoverable reserves below 50 MMBOE and breakeven prices exceeding \$40/bbl., have historically been deprioritized due to their limited hydrocarbon volumes and typically high development costs. However, in the context of energy transition, resource optimization, and the need to maximize production from existing reserves, marginal fields are increasingly being revisited as viable contributors to energy supply.

Objectives and Scope

The objectives of this paper are as follows:

- Systematically identify and analyze the principal technical and economic constraints associated with the development of surface facilities in marginal oil and gas fields.
- Assess advanced, cost-efficient surface facility design methodologies that maximize resource recovery and operational efficiency.
- Demonstrate the application of digitalization, automation, and advanced analytics in minimizing operational expenditures (OPEX) and enhancing system integrity and reliability.
- Highlight optimized fiscal frameworks and contractual models that effectively allocate risk and improve project financial viability.
- Establish pathways for achieving regulatory compliance and environmental sustainability without compromising project economics or performance metrics.

Leveraging empirical insights from the Haliba Field development under Al Dhafra Petroleum (ADNOC), this framework delivers a suite of scalable, adaptable solutions tailored for diverse marginal field scenarios across varying geological and operational contexts.

Key Challenges in Marginal Field Development

Marginal oil and gas fields with limited reserves, challenging geologies, or economically uncertain prospects represent both a significant opportunity and a complex challenge for the energy industry. As global demand for hydrocarbons persists amidst mounting calls for environmental responsibility, the ability to develop marginal fields efficiently has become a critical differentiator. However, the path from discovery to production is fraught with difficulties: technical, financial, environmental, and regulatory. The following are the key challenges inherent to marginal field development:

- **Low Reserves and Declining Flow Rates:** Typically associated with uneconomical payback periods and difficulty in securing investment. These constraints demand ultra-lean capital investment and rapid monetization strategies.
- **High development costs:** Due to lack of economies of scale.
- **Complex geology:** Heterogeneous reservoirs, high water cut, or challenging fluid properties.
- **Remote or harsh environments:** Lead to increased logistical complexity and high cost of infrastructure development, often in geologically and politically sensitive environments.
- **Inadequate Infrastructure:** Absence of nearby pipelines, utilities, and processing facilities demands creative, often costly, solutions that may include mobile or temporary setups.
- **Strict Regulatory Requirements:** Increasing scrutiny over emissions, water use, and environmental impact necessitates cleaner and more efficient designs, which can raise CAPEX if not properly planned.

Addressing these challenges requires a departure from conventional development practices and adoption of agile, cost-efficient methodologies that balance technical robustness with economic pragmatism.

Lean Engineering Strategies for Surface Facilities

Marginal fields often feature complex geology, thin productive zones, and variable reservoir quality. These characteristics make them especially sensitive to suboptimal development decisions. Costs must be rigorously controlled, and every barrel of recoverable resources counts. As a result, conventional approaches to field development may not deliver the efficiency or effectiveness required for commercial success in these settings. Engineering Optimization covers a variety of methods and technologies, from reservoir characterization to facilities design and Operations and it continues to evolve. Lean engineering strategies for marginal field surface facilities emphasize cost reduction and value creation by using optimized designs, streamlining processes, and utilizing existing infrastructure. The following section outlines several critical strategies for surface facilities design tailored to marginal field development:

Minimized Site Development

Minimizing site development is a critical strategy in marginal field projects, where cost efficiency, environmental stewardship, and operational performance must be tightly managed. Utilizing techniques such as minimized grading, gravel pad deployment, and reduced land clearing, operators can significantly lower site preparation costs while enhancing project sustainability. These lean approaches not only reduce environmental disturbance by preserving natural contours and minimizing earthmoving but also accelerate construction timelines. The result is improved project economics through faster mobilization, reduced material usage, and streamlined execution, making them especially valuable for marginal field developments.

Table 1—Summarize Key Strategies and Considerations for Grading, Pad Design in Remote Well Area

Category	Strategy	Considerations
Pad Layout and Design Optimization	Compact Layouts	Use radial or staggered well patterns to reduce site footprint and infrastructure sprawl.
	Terrain Aligned Design	Align well pads with natural land contours to minimize earthmoving and preserve topography.
	Shared Infrastructure	Utilize existing roads, pipelines, and multi-well pads to reduce new construction and grading.
Effective Earthwork Practices	Balanced Cut and Fill	Reuse excavated material onsite to lower hauling costs and environmental impact.
	Early Soil Testing	Conduct geotechnical investigations early to ensure proper compaction and avoid overfilling.
	Minimized Fill Depth	Limit fill volumes to prevent settlement issues and reduce material usage.
Advanced Grading and Stabilization Techniques	Slope Protection	Apply terracing, hinge zones, and geosynthetics to stabilize slopes with minimal grading.
	Erosion Control	Use silt fences, diversion channels, and revegetation to prevent soil loss and maintain integrity.
Environmental and Community Considerations	Sensitive Area Avoidance	Site developments away from wetlands, water bodies, and habitats to protect ecosystems.
	Restoration Planning	Design for future reclamation, enabling rollback of grading and replanting with native species.
	Noise and Visual Mitigation	Use berms and natural features to reduce visual and acoustic impacts on nearby communities.

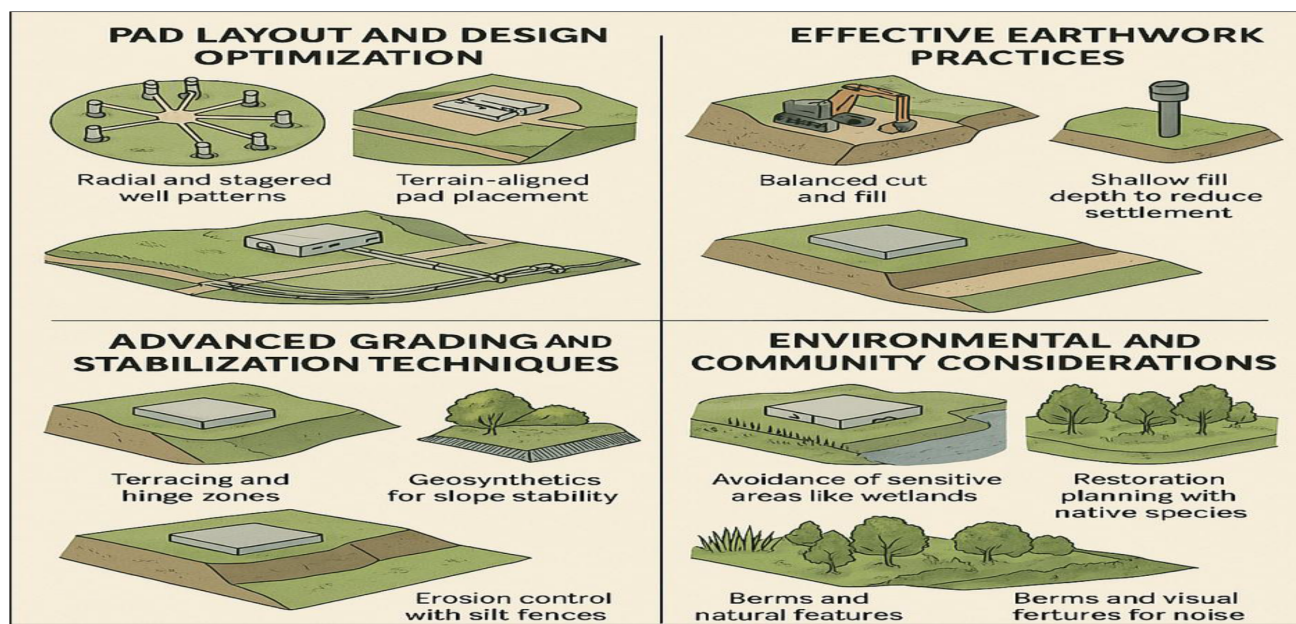


Figure 1—Pictorial view of Site Grading techniques for Remote Well Pads

Comprehensive geotechnical evaluations and the strategic use of temporary structures play a pivotal role in lean site development. Early soil investigations ensure proper compaction, reduce the risk of overfilling or settlement, and support the design of stable, cost-effective foundations. Meanwhile, temporary facilities such as modular offices, laydown areas, and access roads can be rapidly deployed and demobilized, minimizing long-term land disturbance and capital investment. Together, these practices uphold safe operating conditions while maintaining tight control over project costs and timelines.

Efficient Production Gathering Planning

Well Placement Strategies. A well placement strategy determines where, how many, and in what orientation wells are drilled to best exploit the available reserves. This is far more than a technical exercise; it is a comprehensive optimization task that directly impacts field performance, longevity, and economic return. Production well placement strategies can significantly impact the operational efficiency, environmental footprint, and economics of oil and gas developments. Two common configurations are **well clusters** and **scattered wells**, each offering distinct advantages and trade-offs depending on the project context.

Well clusters involve grouping multiple wells in a compact area, often around a central processing or collection facility. This arrangement has several benefits: it minimizes the surface footprint of drilling operations, reduces the length and cost of surface pipelines, and simplifies logistical support such as road access, power supply, and environmental monitoring. For projects in sensitive or densely populated regions, clustering wells can help limit land disturbance and make environmental management more straightforward. Additionally, centralized infrastructure in clustered developments can lead to improved safety oversight and more efficient resource allocation for maintenance and emergency response.

In contrast, scattered well placement involves distributing wells over a broader area, typically positioning each well closer to its target within the reservoir. This configuration can maximize reservoir contact and recovery efficiency, especially in fields where geology is irregular or the reservoir is compartmentalized. The main drawback of scattered wells is the increased surface impact: roads, pipelines, and facilities must be extended over a wider area, leading to higher costs for construction, maintenance, and environmental mitigation. This approach can also complicate logistical support and regulatory compliance, especially in remote or ecologically sensitive regions.

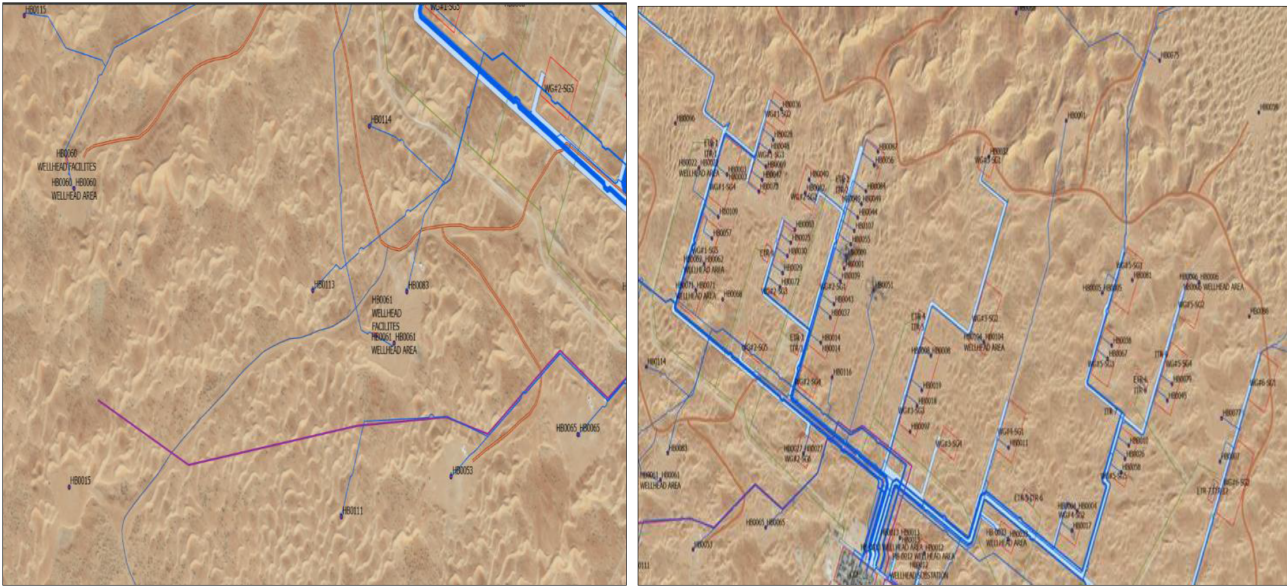


Figure 2—Example for Scattered vs. Grouped Well Clusters at Remote Area

Table 2—Summarize the key benefits between Well Clusters and Scatters Wells arrangement

Aspect	Well Clusters	Scattered Wells
Definition	Grouping multiple production wells on a single pad or in close proximity for centralized operations and shared infrastructure.	Distributing wells across a broader area to maximize reservoir contact and adapt to complex geological conditions.
Surface Impact	Minimized surface disturbance; ideal for sensitive or restricted-access areas.	Larger environmental footprint due to dispersed land use.
Infrastructure	Shared roads, flowlines, power, and processing units reduce CAPEX and OPEX.	Requires extensive infrastructure: longer pipelines, more roads, and dispersed utilities.
Operational Efficiency	Centralized access simplifies monitoring, maintenance, and interventions.	Field operations are more complex due to geographic dispersion.
Reservoir Management	May enhance sweep efficiency in channelized or compartmentalized reservoirs.	Greater reservoir coverage across heterogeneous formations.
Interference Risk	Higher risk of pressure or production interference due to close spacing.	Reduced interference risk with increased well spacing.
Scalability	Limited flexibility for future expansion or reconfiguration.	Easier to adapt and expand based on evolving geological insights.
Capital Requirements	High initial investment for centralized infrastructure.	Potentially lower upfront costs but higher long-term infrastructure expenses.

The optimal placement strategy is **highly dependent on field-specific conditions**, including:

- Reservoir heterogeneity
- Surface accessibility and environmental constraints
- Existing infrastructure
- Economic drivers and project scale
- Regulatory frameworks

A **hybrid approach** using **clustered wells in core areas** and **scattered wells on the periphery** often offer a balanced solution. By understanding and leveraging the unique advantages of each method, operators

can design well placement strategies that balance operational efficiency, environmental stewardship, and economic returns, ultimately supporting more sustainable and effective resource development.

Flowline Cost Minimization Strategies. In marginal fields, even minor bottlenecks can have outsized impacts on production rates due to the narrow margin for economic success. Flowline optimization directly influences the efficiency with which hydrocarbons are transported from the wellhead to processing facilities. A well-optimized flowline system minimizes restrictions and turbulence, ensuring that oil, gas, and associated fluids move smoothly through the system. This efficiency translates to higher well deliverability, allowing operators to maximize daily production rates without incurring excessive operating expenses. Efficient flowline design and operation are essential for reducing both capital expenditures (CAPEX) and operating expenditures (OPEX). A well-executed strategy not only improves flow assurance and system reliability but also contributes to long-term asset integrity and sustainability. The following are key approaches to minimizing flowline costs:

Design Optimization

- **Optimal Diameter Selection:** Carefully balance frictional pressure losses against material and installation costs to determine the most cost-effective pipe diameter. Oversizing increases material costs and high liquid hold-up which can lead to increased corrosion and slugging, while under sizing leads to higher energy consumption and flow inefficiencies.
- **Minimization of Bends and Fittings:** Reducing the number of bends and directional changes in the flowline layout lowers pressure drop, mitigates erosion and corrosion risks, and simplifies maintenance operations.
- **Utilization of Gravity:** Where topography permits, design flowlines to take advantage of natural elevation changes to reduce or eliminate the need for pumping, thereby lowering energy costs.
- **Thermal Insulation:** Apply insulation to maintain fluid temperature within design limits, particularly in lines transporting waxy or hydrate-prone fluids. This helps prevent flow assurance issues and reduces the need for chemical injection or heating systems.

Operational Efficiency

- **Flowline Consolidation:** Combine production from multiple nearby wells into shared flowlines (Trunklines or Gathering lines) to reduce the total number of lines required. This approach minimizes surface disturbance and infrastructure redundancy.
- **Real-Time Monitoring Systems:** Implement sensors and telemetry to continuously monitor pressure, temperature, and flow rates. Early detection of anomalies enables proactive maintenance and reduces downtime.
- **Dynamic Flow Control:** Adjust flow rates based on reservoir performance and equipment capacity to optimize throughput, reduce wear on mechanical components, and enhance energy efficiency.
- **Preventive Maintenance Programs:** Establish regular pigging schedules, corrosion monitoring, and integrity inspections to extend flowline service life and prevent unplanned failures.

Cost Control Measures

- **Resource Optimization During Construction:** Streamline construction workflows to minimize material wastage, labour inefficiencies, and rework. Use modular construction techniques where feasible.
- **Strategic Procurement:** Leverage bulk purchasing agreements and prioritize local sourcing to reduce lead times and procurement costs.

- **Corrosion Management:** Employ protective coatings, corrosion-resistant alloys, or cathodic protection systems to mitigate internal and external corrosion risks, especially in severe corrosive service environments.
- **Adoption of Advanced Technologies:** Explore innovative solutions such as anti-scale and anti-corrosion devices, smart coatings, and self-cleaning flowline technologies to enhance performance and reduce maintenance.
- **Life Cycle Cost Analysis:** Evaluate total cost of ownership including installation, operation, maintenance, and decommissioning when selecting materials and design configurations to ensure long-term cost efficiency.
- **Integrated Production Planning:** Align well deliverability forecasts with flowline capacity to avoid underutilization or overloading, ensuring optimal system performance and investment return.



Figure 3—Representation of optimized Flowlines gathering network in Remote Fields

Well placement and flowline design are inextricably linked. A cohesive approach ensures that wells are sited within reach of cost-effective, reliable flowline networks, further enhancing project economics and operability. Poor alignment between well locations and flowline infrastructure can lead to excessive costs, operational complexity, or compromised production. Minimizing flowline costs requires a holistic approach that integrates smart design, efficient operations, and strategic cost control. By optimizing pipe sizing, leveraging terrain, consolidating infrastructure, and adopting advanced monitoring and corrosion mitigation techniques, operators can significantly reduce both upfront and lifecycle expenditures.

Multiphase Pumping Technology. Unlike major fields with robust pressure support and straightforward production profiles, marginal fields often struggle with weak reservoir pressure and mixed-phase flow. This means that oil, gas, and water exit the wellbore together in varying proportions a phenomenon known as multiphase flow. The complexities of handling such mixtures can lead to operational difficulties, increased costs, and reduced recovery, particularly when conventional equipment is not designed to deal with multiphase streams.

Traditionally, production from oil and gas wells is routed to a gathering station where the fluids are separated. This requires the construction of surface facilities such as separators and associated pipelines for oil, gas, and produced water. In marginal fields, setting up this infrastructure may not be economically justified due to the limited volume of recoverable hydrocarbons. Furthermore, conventional pumps are

designed to handle liquid or Compressors for gas and will perform poorly or even fail when presented with a mixture of oil, gas, and water.

Multiphase pumping systems are engineered to simultaneously transport oil, gas, and water through a single pipeline, eliminating the need for phase separation at the wellsite. They can handle varying proportions of each phase, adjust to changing well conditions, and transmit the entire production stream through a single pipeline to a central processing facility, sometimes located many kilometres away. This technology is particularly transformative in remote, and subsea fields, where conventional infrastructure is either cost-prohibitive or logistically challenging. By integrating flow management across phases, multiphase pumps deliver substantial economic, operational, and environmental advantages.

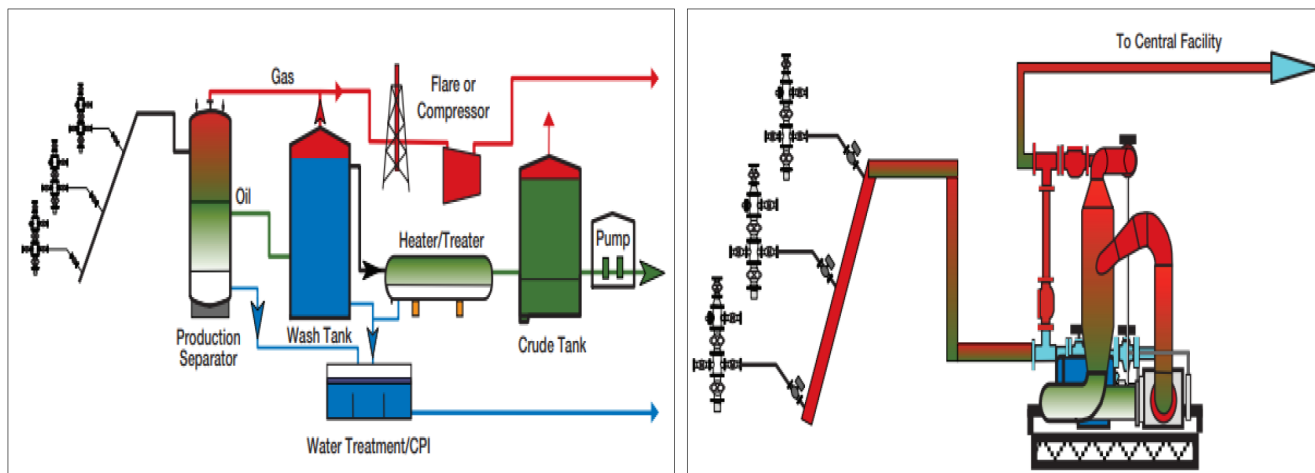


Figure 4—Representation of conventional Separation versus Multiphase Pumping System

Cost Reduction Benefits

- **Reduced Surface Infrastructure:** Multiphase systems eliminate the need for traditional separation and treatment facilities such as separators, dehydrators, and compressors. This reduction in surface equipment minimizes site complexity and footprint, especially in environmentally sensitive areas.
- **Lower Capital Expenditures (CAPEX):** Compared to installing discrete processing units, multiphase pumps offer a more economical solution by consolidating functionality into a single system. This is particularly beneficial in offshore and marginal field developments.
- **Reduced Operating Costs (OPEX):** With fewer mechanical components and surface facilities, operators benefit from lower maintenance requirements, reduced energy consumption, and streamlined staffing needs.
- **Accelerating Production Start-up:** Deploying multiphase pumps can reduce project timelines by allowing early production without waiting for full-scale separation facilities to be built. This accelerates cash flow and improves project economics.
- **Extended Economic Life of Fields:** Multiphase pumping enables continued production from low-pressure or declining wells by overcoming backpressure constraints, thereby enhancing recovery and prolonging field viability.

Operational Advantages

- **Enhanced Production Rates:** By lowering wellhead pressure and mitigating backpressure effects, multiphase pumps facilitate greater drawdown and improved flow rates, maximizing reservoir productivity.

- **Improved Flow Assurance:** These systems help manage complex flow regimes, including slugging, hydrate formation, and terrain-induced flow instabilities. This ensures more consistent and reliable production, particularly in deepwater and hilly terrains.
- **Reduced Environmental Impact:** The compact design and centralized processing reduce land use and emissions, aligning with sustainability goals and regulatory compliance.
- **Well Stimulation Effects:** Multiphase pumping can disrupt hydrate formation and reduce reliance on chemical inhibitors such as methanol, simplifying flow assurance strategies and lowering chemical handling risks.

Strategic Applications

- **Remote and Subsea Fields:** In deepwater and isolated locations, multiphase pumps eliminate the need for extensive topside infrastructure, enabling economic development of otherwise stranded assets.
- **Onshore Marginal Fields:** These systems facilitate the tie-in of low-rate or geographically dispersed wells by reducing the need for complex gathering networks and surface facilities.
- **Zero Flaring Initiatives:** By transporting raw gas and liquids to centralized processing hubs, multiphase pumping supports emissions reduction and environmental, social, and governance (ESG) objectives.

While multiphase pumps offer significant benefits for marginal field development such as reducing surface infrastructure and enabling early production their deployment requires careful planning. Key considerations include evaluating fluid properties like viscosity and gas content, ensuring reliable power supply in remote areas, and addressing maintenance challenges through robust design and remote monitoring systems.

In summary, multiphase pumps are not just optional tools they are increasingly fundamental to the viability of marginal field development. By enabling efficient, flexible, and cost-effective transport of mixed-phase fluids, they help unlock reserves that would otherwise remain uneconomic. Their role in minimizing infrastructure, reducing environmental impact, and adapting to the variable nature of marginal fields makes them indispensable as the industry seeks to maximize the value of every barrel.

Lean Oil Processing Configurations

Lean oil processing, which focuses on streamlining and simplifying the treatment and handling of produced fluids, plays a pivotal role in unlocking the value of these marginal resources. Lean oil processing refers to the approach of reducing the complexity, footprint, and cost of processing equipment and procedures in the production of hydrocarbons. Instead of traditional, equipment-heavy facilities that separately treat oil, gas, and water, lean processing uses integrated systems to handle multiphase fluids more efficiently. This methodology is especially suited for marginal fields, where investment returns are tightly constrained.

Selecting the appropriate oil processing configuration is a critical decision in marginal field development, where economic constraints, infrastructure limitations, and crude quality must be carefully balanced. The choice of configuration directly influences safety, marketability, and operational feasibility. The three primary approaches **no stabilization**, **partial stabilization**, and **full stabilization** offer distinct trade-offs in terms of cost, complexity, and product quality.

No Stabilization (Basic Separation). This configuration involves the basic separation of well fluids into oil, gas, and water without further treatment to reduce crude volatility. It is typically employed in early production systems or pilot phases.

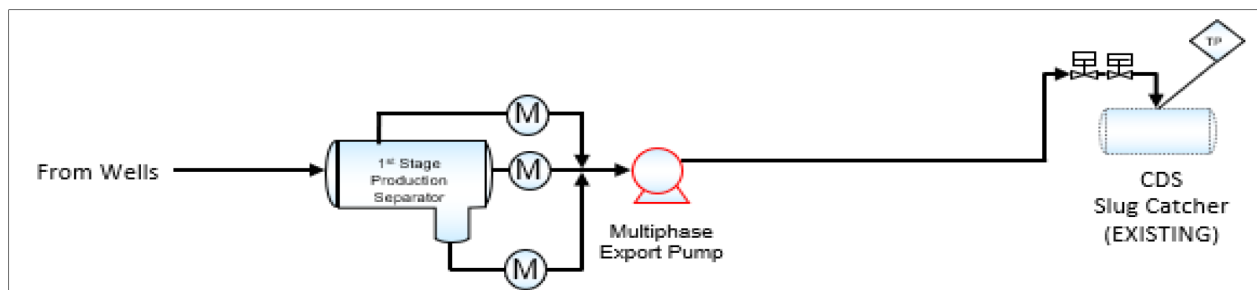


Figure 5—Process Schematic for Basic Oil Separation (No Stabilization or Full Well Stream transfer)

Advantages:

- Minimal capital and operating expenditures.
- Simplified facility design with a small equipment footprint.

Limitations:

- High crude vapor pressure poses safety risks during storage and transport.
- May not meet pipeline or export specifications.
- Unsuitable for long-distance transport or regulatory compliance in many markets.

Partial Stabilization. Partial stabilization removes a portion of volatile components typically through gas stripping or a basic stabilization column to lower the crude's vapor pressure and improve handling characteristics.

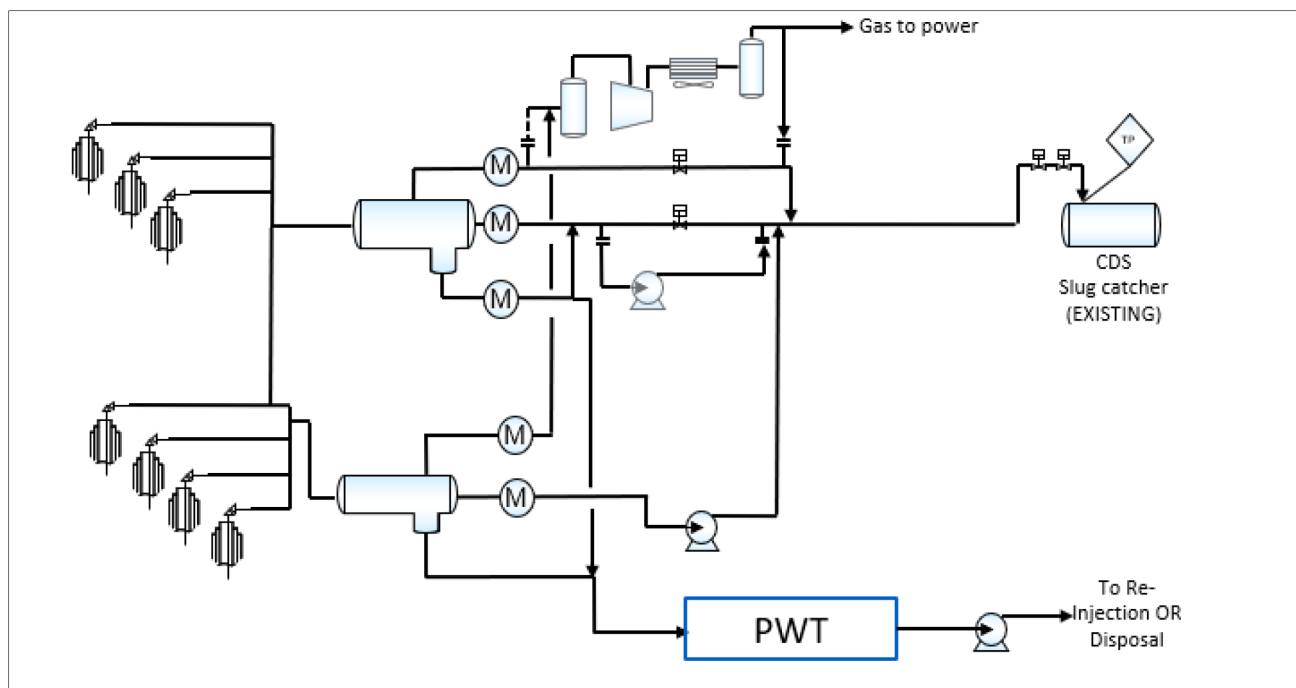


Figure 6—Process Schematic for Partial Stabilization

Advantages:

- Offers a balanced approach between cost and product quality.
- Enhances safety and reduces emissions during storage and transport.
- Minimizes loss of light hydrocarbons, preserving revenue.
- Provides operational flexibility to adapt to changing reservoir conditions.

Ideal for marginal fields where full-scale infrastructure is economically unfeasible, but improved crude quality is necessary for market access or blending.

Full Stabilization. Full stabilization employs heated stabilization columns and condensers to remove most volatile components, producing pipeline- or export-grade crude that meets stringent market specifications.

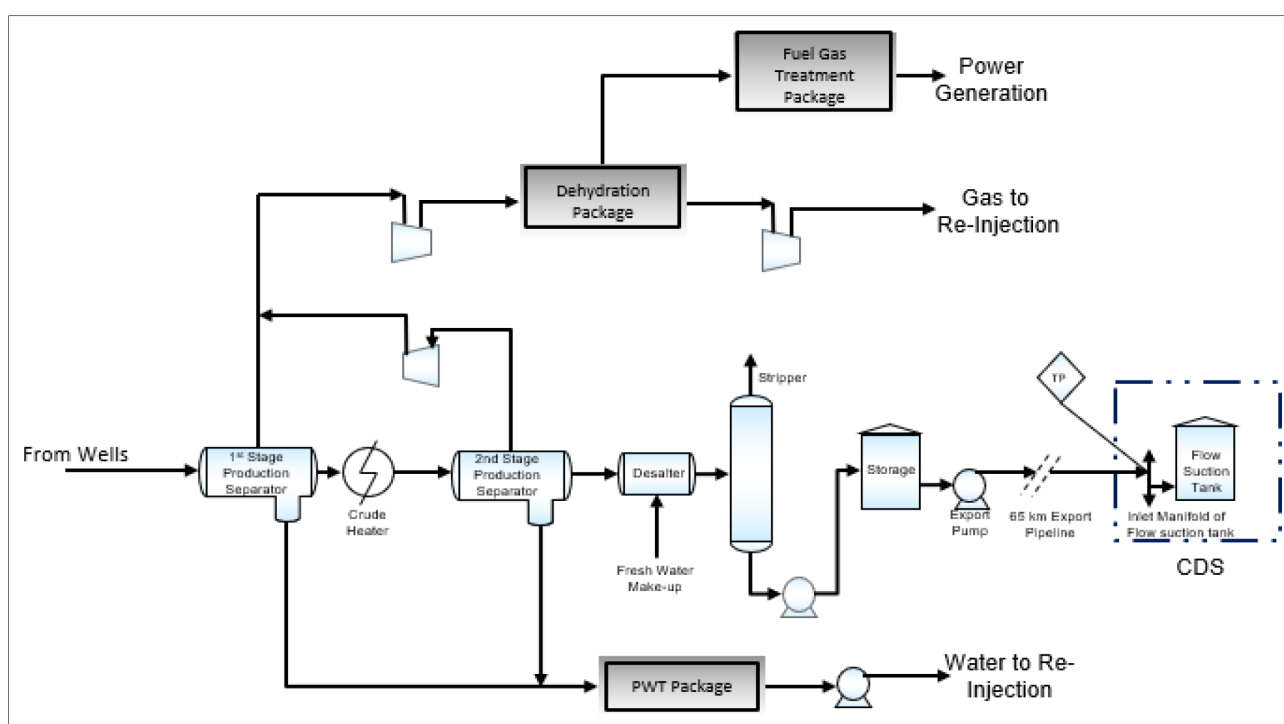


Figure 7—Process Schematic for Full Stabilization

Advantages:

- Delivers high-quality stabilized crude suitable for direct export or pipeline transport.
- Maximizes product pricing and marketability.
- Ensures compliance with regulatory and commercial standards.

Limitations:

- Requires significant capital investment and energy input.
- Larger equipment footprint and operational complexity.
- May be economically unjustifiable for small or declining fields.

Strategic Benefits of Stabilization in Marginal Fields.

- **Cost Efficiency:** Partial stabilization provides a pragmatic solution for small-scale developments, balancing investment with operational needs.
- **Improved Product Handling:** Lower vapor pressure enhances safety, reduces emissions, and simplifies logistics.
- **Revenue Protection:** Retains valuable light ends that would otherwise be lost in basic separation.
- **Regulatory and Market Compliance:** Ensures crude meets specifications for blending, trading, or pipeline entry.

In the context of marginal oil fields, **partial stabilization** emerges as the most effective compromise delivering improved crude quality and operational safety without incurring the full cost and complexity of complete stabilization systems. The selected processing strategy should be tailored to the field's production profile, economic thresholds, and market requirements, ensuring optimal resource utilization and long-term viability.

To maximize efficiency and ensure cost-effectiveness, all process equipment is specified strictly as fit-for-purpose, adhering rigorously to recognized international standards. No excessive design margins or gold-plating are introduced, meaning that every component is engineered to meet operational requirements without unnecessary over-specification. This disciplined approach not only optimizes capital expenditure but also supports sustainable asset performance across the project lifecycle.

To derive maximum value from marginal fields, operators must embrace approaches that ensure every barrel produced is cost-effective, safe, and environmentally responsible. Lean oil processing is not merely a preference but a requirement in these contexts. By minimizing capital and operational expenditures, addressing technical and environmental challenges, and offering unparalleled operational flexibility, lean oil processing paves the way for sustainable hydrocarbon production from resources that would otherwise remain untapped or uneconomic.

Export System Flexibility

An appropriate export strategy is indispensable in marginal field development planning because it is the crucial link between the field and the market. It ensures that every barrel produced can be efficiently, safely, and profitably delivered to buyers. Export strategy refers to the logistical, technical, and commercial blueprint by which produced hydrocarbons are transported from the field to market. This encompasses the selection of export routes (pipelines, trucking, shipping, etc.), transportation infrastructure, sales agreements, regulatory compliance, and risk mitigation related to product evacuation. In marginal fields, where profit margins are inherently slim, any inefficiency or bottleneck in export can rapidly erode economic returns.

Selecting the appropriate export strategy is a critical component of marginal field development planning. Given the typically constrained economics, variable production profiles, and remote locations of such fields, the export method must balance capital investment, operational flexibility, and infrastructure availability. The three primary options **pipelines**, **trucking**, and **tiebacks to existing facilities** each offer distinct advantages and limitations depending on field-specific conditions.

Table 3—Summarize the key benefits among different export systems

Export Method	Advantages	Disadvantages	Best Fit
Pipelines	- Economical for high production volumes - Enables continuous, reliable flow - Minimal manual handling	- High upfront capital cost (CAPEX) - Inflexible if production declines or shifts	- Fields with sustained high volumes - Locations with access to existing pipeline corridors or central facilities
Trucking	- Low initial investment - High flexibility for variable rates - Accessible for remote locations	- Higher operating cost (OPEX) - Vulnerable to weather and road conditions which pose HSE Risk - Logistical delays possible	- Small-scale or early production fields - Remote locations without pipeline access - Temporary export during appraisal or EPF phases
Tiebacks to Existing Facilities	- Leverages existing infrastructure - Reduces CAPEX - Accelerates development timelines	- Requires host facility capacity assessment - May need technical modifications	- Fields near underutilized or expandable processing hubs - Projects with limited standalone viability

Export strategy selection for marginal fields must be guided by a careful evaluation of production scale, infrastructure availability, and economic thresholds:

- **Pipelines** are optimal for large, long-life developments with stable output.
- **Trucking** provides agility and cost control for early-stage, remote, or low-volume operations.
- **Tiebacks** offer a cost-effective solution when compatible infrastructure is nearby.

In the absence of dedicated pipeline infrastructure, marginal field operators can deploy interim solutions such as trucking, mobile export systems, or temporary tie-ins.

While sharing export facilities with joint ventures (JVs) can reduce costs and accelerate deployment, it introduces operational and commercial risks. Dependencies on third-party schedules, priorities, and performance can lead to unplanned downtime or restricted access. Disputes, misaligned maintenance plans, or regulatory breaches by one partner may impact all users. To mitigate these risks, operators must establish clear usage rights, robust contracts, transparent communication protocols, and contingency plans ensuring operational flexibility and protecting revenue streams in the volatile context of marginal field development.

These approaches enable early cash flow generation, defer major capital commitments, and allow field performance validation before scaling up. Integration with downstream facilities or shared logistics networks can further enhance cost efficiency and operational resilience.

Digitalization & Automation

In the fast-evolving landscape of oil and gas, marginal field development demands relentless focus on efficiency, cost control, and operational agility. Digitalization and automation have become indispensable tools in this pursuit, offering solutions to the unique challenges that marginal fields present. Digital technologies enable real-time monitoring, predictive maintenance, and advanced analytics, allowing operators to swiftly identify inefficiencies and make data-driven decisions. Automation, meanwhile, reduces manual intervention, streamlines workflows, and ensures that operations run seamlessly around the clock.

The Facilities are engineered as autonomous, standalone assets, operating independently from both the remote facility and centralized processing control rooms. This design reflects a strategic shift toward minimal manning, full automation, and remote operability, aligning with operators' broader digital transformation and operational efficiency goals.

Maintenance and Reliability Strategy

The maintenance philosophy is built around **predictive analytics**, **remote diagnostics**, and **campaign-based interventions**, minimizing site presence while maximizing reliability:

- **Online Condition Monitoring:** Real-time surveillance of rotating and static equipment using smart sensors and analytics platforms to detect anomalies and degradation trends.
- **Smart Predictive Maintenance Tools:** AI-driven systems forecast equipment health and schedule interventions proactively, reducing unplanned downtime.
- **Self-Diagnostic Instrumentation:** Intelligent field devices continuously report operational status, fault conditions, and calibration needs, enabling autonomous diagnostics.
- **Campaign-Based Maintenance:** Maintenance activities are grouped and executed periodically via third-party contracts, reducing operational overhead and concentrating resources at headquarters.

Integrity and Corrosion Management

A robust **Integrity Management System (IMS)** is deployed to ensure long-term asset reliability and regulatory compliance. Key components include:

- **Corrosion Mitigation Systems:** Applied across wells, pipelines, and surface facilities, these systems incorporate chemical injection, corrosion-resistant materials, and real-time monitoring technologies.
- **Risk-Based Inspection (RBI) and Fitness-for-Service (FFS) Assessments:** These methodologies prioritize inspection and maintenance activities based on risk profiles and operational criticality, ensuring efficient resource allocation and compliance with ADNOC standards.

This design exemplifies a modern, autonomous operational model tailored for efficiency, safety, and reliability. Through full automation, remote control, predictive maintenance, and integrity-focused design, the facility minimizes on-site presence while maximizing performance and lifecycle value. With minimal on-site staff, operators can achieve high levels of process control, emissions monitoring, and regulatory compliance. In essence, the synergy of digitalization and automation transforms marginal field development from a high-risk, high-cost endeavor into a streamlined, resilient, and economically attractive opportunity.

Smart Fiscal Metering Strategies

Fiscal metering is indispensable for marginal fields due to their unique economic sensitivities and operational challenges. Unlike large reservoirs, marginal fields operate with tight profit margins and shorter lifespans, making every drop in production and every dollar of revenue crucial to overall viability.

Accurate fiscal metering ensures precise measurement of produced hydrocarbons at key custody transfer points, forming the backbone of transparent revenue calculation, taxation, and royalty allocation. For marginal fields, even small discrepancies in measurement can lead to significant financial losses, disputes, or regulatory penalties, swiftly eroding profitability.

In marginal field developments, fiscal metering must strike a balance between **measurement accuracy**, **cost-efficiency**, and **regulatory compliance**. These fields often present unique challenges such as **multiphase flow**, **remote locations**, and **limited infrastructure**, requiring tailored metering solutions that are both technically sound and economically viable.

Metering Technologies

- **Multiphase Flowmeters (MPFMs):**
MPFMs are particularly useful in early production phases or where separation is impractical. While they may not meet the highest standards for fiscal accuracy, they provide valuable real-time

data for production monitoring and allocation. Their compact design and ability to handle mixed-phase flow make them suitable for remote or space-constrained installations.

- **Single-Phase Metering Post-Stabilization:**

Where partial separation or stabilization is feasible, single-phase metering using custody transfer-grade meters becomes viable. These meters offer high accuracy and are typically compliant with fiscal and regulatory standards, making them ideal for final sales measurement.

- **Meter Provers:**

Critical for validating custody meters during commissioning and routine operations, meter provers ensure traceable, repeatable measurements. Their use is essential for maintaining confidence in reported volumes and avoiding fiscal disputes.

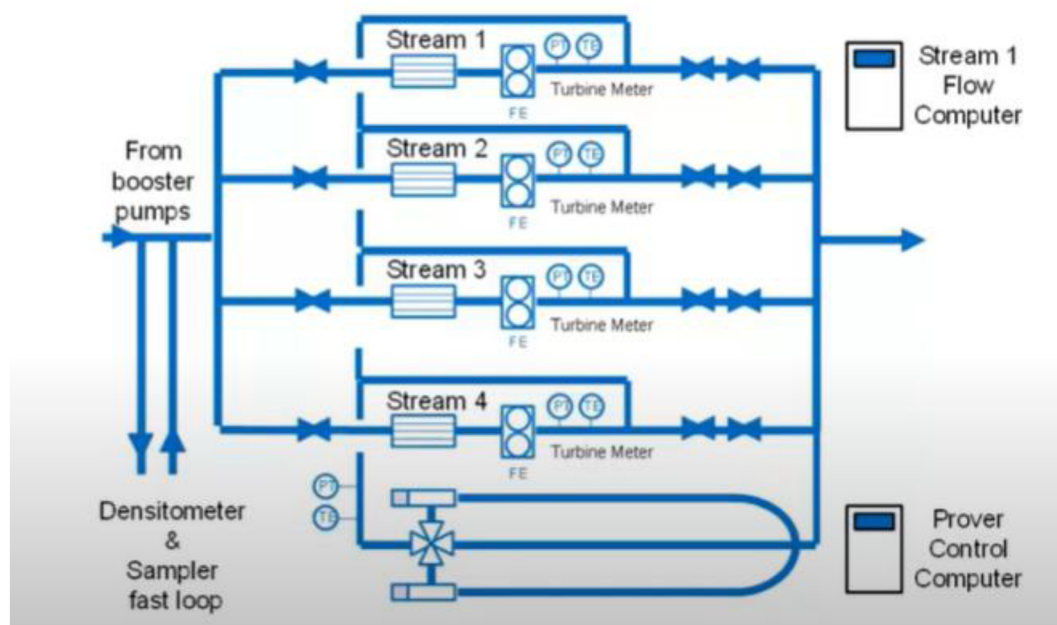


Figure 9—Example for Oil Custody Transfer Meter Package Schematic

Table 4—Outlining Meter Accuracy and Suitability for Upstream Oil and Gas Applications

Meter Type	Accuracy ($\pm\%$)	Suitability for Upstream Applications	Key Advantages	Limitations
Coriolis Meter	0.1% - 0.5%	Multiphase flow, wellhead monitoring, custody transfer	High accuracy, measures mass flow & density, unaffected by flow profile	High cost, sensitive to vibration
Ultrasonic Meter (Clamp-on)	1% - 5%	Non-intrusive monitoring, leak detection, pipeline flow	No pressure drop, portable, works on existing pipes	Lower accuracy, affected by gas bubbles/solids
Turbine Meter	0.5% - 1%	Single-phase oil/gas flow, fiscal metering	High accuracy for clean fluids, proven technology	Sensitive to wear requires clean fluids
Orifice Plate Meter	1% - 2%	Gas flow measurement, well testing	Simple, low cost, widely accepted	Pressure loss, requires straight pipe runs
Vortex Meter	1% - 1.5%	Gas & steam flow, flare gas measurement	No moving parts, handles high pressure/temperature	Limited to single-phase, affected by pulsations
Positive Displacement (PD) Meter	0.2% - 0.5%	Crude oil measurement, fiscal metering	High accuracy for viscous fluids	Sensitive to wear requires filtration
Multiphase Flow Meter (MPFM)	5% - 10%	Well testing, production optimization	Eliminate separators, real-time data	High cost, complex calibration

Fiscal Allocation Practices

- **Shrinkage Factors:** When metering partially stabilized crude, shrinkage factors must be applied to account for gas evolution and convert measured volumes to stock tank conditions. These factors should be supported by laboratory analysis or field data to ensure defensibility.
- **Validation and Calibration:** Regular calibration and system checks are essential, especially under dynamic flow conditions. These practices uphold measurement integrity and ensure compliance with contractual and regulatory obligations.

Field-Specific Challenges

- **Remote Locations:** Metering systems in remote areas must be designed for minimal maintenance. Solutions may include modular skid-mounted units, remote diagnostics, and automated validation tools to reduce field visits and operational overhead.
- **Limited Infrastructure:** The choice of metering technology is often constrained by available utilities such as power, separation facilities, and communication systems. Low-power, standalone systems may be necessary in such contexts.
- **Complex Fluids:** High gas-oil ratios (GORs), variable water cuts, and emulsion behaviour demand metering systems with advanced flow algorithms and configurable settings to maintain accuracy across changing conditions.

Economic Viability

- **Capex/Opex Optimization:** Metering solutions should be selected based on lifecycle cost analysis, ensuring that operational expenses remain low while meeting fiscal standards. Modular and scalable systems can help defer capital investment until production performance is validated.
- **Cash Flow Modelling:** Financial models must incorporate metering uncertainty, tax obligations, and production forecasts to assess project viability. Sensitivity analysis around measurement accuracy can help identify risk thresholds and guide investment decisions.

A robust fiscal metering strategy for onshore marginal fields integrates **fit-for-purpose technologies**, **validated allocation practices**, and a comprehensive understanding of **regulatory and economic frameworks**. By aligning metering systems with field conditions and fiscal requirements, operators can ensure **measurement integrity**, **regulatory compliance**, and **economic sustainability** throughout the project lifecycle.

Tailored Execution Strategy

Developing onshore marginal fields requires a **phased, cost-conscious, and adaptable execution strategy** that accelerates early production while managing financial exposure. The approach must be tailored to field-specific conditions, leveraging modular and scalable solutions to optimize capital efficiency and operational flexibility.

Table 5—Summarize the key execution approaches to be considered in Surface Facilities Design

Category	Strategy	Description
Phased Development	Pilot / Temporary Systems	Deploy small-scale EPS to validate reservoir performance, enable early cash flow, and reduce financial risk.
	Phased Expansion	Scale up production using modular units or permanent infrastructure based on pilot results; defer major CAPEX until justified.
Modular & Standardized Design	Prefabricated Modules	Use skid-mounted units for processing and storage to minimize site work and simplify logistics.
	Standardized Components	Streamline procurement and spares management; enhance compatibility across developments.
Early Production Systems (EPS)	Rapid Deployment	Enable quick installation and early revenue generation; support performance monitoring.
	Scalability	Expand or redeploy EPS based on production trends; transition into permanent facilities if needed.
Fit-for-Purpose Solutions	Tailored Facility Design	Align systems with fluid properties and flow profiles; avoid over-engineering.
	Cost-Effective Engineering	Apply lean design principles to balance performance, reliability, and lifecycle cost.

Strategic Benefits

This execution model delivers several key advantages:

- **Accelerated time to first oil** through early production systems.
- **Deferred major CAPEX**, reducing financial risk.
- **Operational flexibility** to adapt to reservoir behaviour.
- **Improved project economics** through lean, scalable design.
- **Reduced exposure** by validating assumptions before full-scale investment.

A phased, modular, and fit-for-purpose development strategy is essential for unlocking the potential of onshore marginal fields. It enables operators to manage uncertainty, optimize capital deployment, and transition efficiently from exploration to sustained production all while maintaining technical integrity and economic viability.

Contracting Strategies Framework

Selecting the optimal project delivery strategy for marginal field developments is a nuanced decision that hinges on balancing cost, risk, and operational control. Each model EPC, DBO, and O&M service contracts offers distinct advantages and trade-offs that must be aligned with the field's complexity and the operator's strategic priorities.

The **Engineering, Procurement & Construction (EPC)** model is a turnkey approach where a single contractor assumes full responsibility for design, procurement, construction, and commissioning. This structure is particularly effective for projects with well-defined scopes and moderate complexity. Its key strength lies in offering cost and schedule certainty through fixed pricing and integrated execution. However, this certainty comes at a premium, as contractors typically embed risk contingencies into their pricing. Additionally, EPC contracts are rigid; any scope changes post-award can lead to significant cost escalations and delays. For operators seeking minimal involvement and rapid deployment, EPC remains a reliable choice.

In contrast, the **Design-Build-Operate (DBO)** model extends the contractor's role beyond construction into long-term operations. This strategy is performance-driven, incentivizing the contractor to optimize design for operational efficiency and maintain high standards throughout the asset's lifecycle. DBO is particularly suited for marginal fields where sustained performance and cost control are critical. However, the complexity of structuring DBO contracts especially around KPIs, risk-sharing, and dispute resolution requires careful planning. Operators must ensure that roles and expectations are clearly defined to avoid friction during the operational phase.

Operations & Maintenance (O&M) service contracts offer a more flexible and modular approach. These contracts are typically used post-construction to manage day-to-day operations and maintenance. They are ideal for transitional phases or when the operator retains ownership and strategic control but outsources execution. O&M contracts allow for adaptability in response to changing field conditions and are often structured with buy-back or demolition clauses, especially in temporary installations like produced water treatment units. While they offer cost control and operational continuity, they may lack the integrated performance incentives found in DBO models.

Ultimately, the choice between EPC, DBO, and O&M should be guided by the operator's risk appetite, desired level of involvement, and long-term strategic goals. Marginal fields, by nature, demand lean execution and high efficiency, making hybrid models or phased strategies increasingly attractive. For instance, an EPC phase followed by an O&M contract can balance rapid deployment with operational flexibility, while a DBO model may be preferable for fields requiring sustained performance and lifecycle cost optimization.

Effective Risk Management Strategies

Managing marginal fields effectively requires embedding risk mitigation strategies from the earliest stages of project planning. These fields, characterized by lower reserves and shorter production lifespans, demand a design philosophy that prioritizes flexibility, cost control, and end-of-life value recovery.

Table 6—Summarize the Risks Management Strategy and Benefits in Marginal Field Development

Risk Management Strategy	Key Elements	Benefits
1. Easy Decommissioning by Design	- Modular, skid-mounted units for phased removal - Temporary foundations to reduce remediation - Plug-and-play utility connections - Accurate as-built documentation and asset tagging	- Lower abandonment costs - Faster site clearance - Simplified dismantling and relocation
2. Enhancing Resale or Reuse Value	- Use of standardized, re-deployable equipment - Skid-mounted/tailorized units for mobility - OEM-supported designs with resale programs	- Increased marketability - Higher resale potential - Continued serviceability
3. Contractual & Commercial Flexibility	- Short-term lease or O&M contracts - Buy-back agreements with vendors - Predefined decommissioning triggers and financial safeguards	- Reduced sunk costs - Clear exit strategies - Financial risk containment
4. Regulatory and Environmental Risk Mitigation	- Early engagement with regulators and communities - Low-impact site development - Periodic decommissioning readiness audits	- Avoidance of delays and penalties - Lower remediation costs - Improved compliance
5. Financial Risk Controls	- Life-cycle cost modelling including abandonment - Insurance for decommissioning liabilities - Salvage value recovery in financial planning	- Improved project economics - Risk transfer and protection - Enhanced bankability

Ultimately, effective risk management transforms potential vulnerabilities into strategic advantages. It provides a structured approach for navigating the unique challenges of marginal fields, enabling operators to maximize returns, uphold regulatory and environmental standards, and maintain financial resilience throughout the project's life cycle.

Economic Efficiency

Developing marginal fields demands a disciplined economic strategy due to their limited reserves, shorter production horizons, and heightened financial sensitivity. Success hinges on minimizing capital expenditure (CAPEX), controlling operational expenditure (OPEX), and accelerating returns through smart design and execution choices.

Table 7—Strategic Financial Optimization in Marginal Field Economics

Strategy Area	Key Elements	Benefits
1. CAPEX/OPEX Trade-Off Analysis	- Fit-for-purpose, low-spec equipment to reduce CAPEX - Leasing mobile equipment to preserve capital - Energy-efficient systems and automation to reduce OPEX - Shared services and infrastructure	- Lower upfront investment - Faster deployment - Long-term operational savings
2. Phased Investment Approach	- Pilot or early production phase to generate early cash flow - Modular facility design for scalable growth - Deferral of non-critical infrastructure until justified	- Reduced financial exposure - Flexibility to adapt to reservoir performance - Improved capital efficiency
3. Quick Payout Periods	- High-rate early wells targeting productive zones - Use of prefabricated systems or EPFs for fast deployment - Cash flow-aligned contracts and procurement	- Accelerated revenue generation - Shortened time-to-first oil - Improved financial planning
4. Fiscal Regime Impacts	- Leverage incentives like tax holidays and royalty reductions - Sensitivity analysis for robustness under fiscal scenarios - Ring-fencing to ensure cost recovery within the field	- Enhanced project economics - Protection against fiscal volatility - Increased investor confidence
5. Economic Screening Tools	- Adjusted NPV and IRR for short lifespans - Payback period as a key metric (≤ 3 years) - Breakeven analysis for oil price thresholds	- Informed investment decisions - Clear profitability benchmarks - Risk-adjusted financial modelling

Economic optimization in marginal fields is achieved through low upfront CAPEX, controlled OPEX, phased investments, rapid payback strategies, and strategic use of fiscal incentives. These elements collectively enhance project bankability, reduce financial exposure, and improve the likelihood of long-term success.

Case Study: Haliba Field Development

The Haliba Field, developed by Al Dhafra Petroleum (ADNOC), stands as a benchmark for successful marginal field development through the strategies described above sections. Key outcomes include:

- **CAPEX Reduction:** Achieved a 25% decrease in capital expenditure by employing a phased development strategy, embracing lean engineering design, standardizing processes, modularizing facilities, and utilizing smart fiscal metering techniques.
- **Operational Efficiency:** Operating expenses (OPEX) were cut by 20% through digital integration and the implementation of remote operations, underscoring the value of technology in driving efficiency.
- **Faster Execution:** Project delivery timelines improved by over 30%, by adopting effective contracting strategies, proactive risk mitigation, prefabricated infrastructure, and streamlined regulatory approvals.
- **Environmental Stewardship:** The Haliba Field achieved its emissions targets and reduced land use impact through lean site design and the use of multiphase pumping systems, helping to protect environmentally sensitive areas and maintain sustainability.

The success of the Haliba Field project is attributed to a high degree of alignment and integration across multidisciplinary team including reservoir engineering, technical operations, fiscal management, and regulatory compliance. This collaborative environment enabled the team to proactively identify and address emerging risks, adapt promptly to evolving project parameters, and translate risk management strategies into effective, actionable on-site practices.

The Haliba Field case study demonstrates the efficacy of integrating advanced risk management strategies and technical innovation in marginal field development. Through a disciplined approach encompassing innovation, cross-functional collaboration, and meticulous planning, projects can systematically de-risk operations and unlock substantial value delivering superior outcomes across financial, operational, and environmental domains.

Conclusion

Marginal fields are no longer just stranded assets; they are strategic opportunities in an evolving energy landscape. By combining lean engineering, digitalization, fiscal prudence, and environmental stewardship, operators can unlock sustainable value. The framework presented in this paper, validated by field-proven practices, offers a roadmap for the economic and responsible development of marginal onshore resources. Future work may focus on integrating carbon capture, ESG-aligned financing, and AI-driven reservoir modeling to further enhance marginal field viability in the face of global energy transformation.

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